Slow-controls simulation

Updated VD and HD results; a short guide to running them

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6 October 2026

DEEP UNDERGROUND NEUTRINO EXPERIMENT

Fermilab

Monitoring: two processes per detector

VD AND HD TESTED SEPARATELY

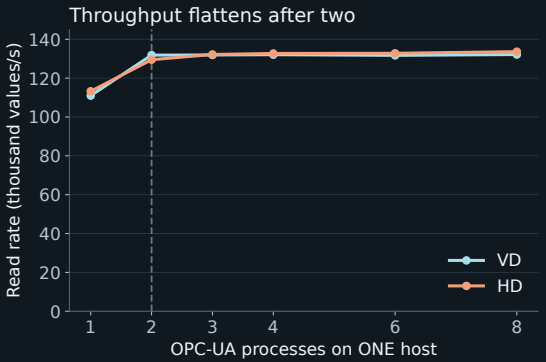
Two processes reached 95% of the best throughput

All 12 process-count checks passed, with zero read errors.

Detector	WIBs	DAPHNEs	Read values/s	Memory
VD	480	42	131,838	296 MB
HD	750	150	129,467	354 MB

[L1] At two OPC-UA processes on one test host. Board-only model: 17 variables/WIB, 21/DAPHNE; PTCs and other systems excluded.

Beyond two processes, throughput changes little



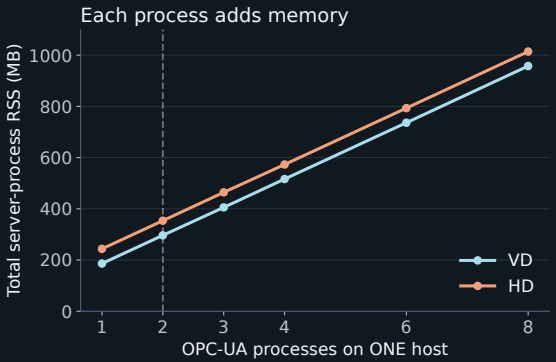


Diagram. Sweep: 1, 2, 3, 4, 6 and 8 OPC-UA programs sharing one test host.

[L1] One representative value/board changes every 250 ms. VD DAPHNE interference fell from +13.1% at two processes to +1.3% at four.

Configuring and checking one WIB is one job

QUEUE MODEL

48 simultaneous WIB jobs per bridge

Each worker configures one board, reads its settings back, then takes the next board.

Detector	Bridges	WIBs/bridge	Batches	p95	p99
VD	4	120	3	**48.42 s**	78.41 s
HD	6	125	3	**62.28 s**	78.41 s
HD comparison	4	187–188	4	78.41 s	94.55 s

[Q1] VD batches: 48 + 48 + 24; HD/six bridges: 48 + 48 + 29. Assumes 16.10 s/WIB, network delays and retries; HD reuses VD timings. Projections, not measured hardware completion.

Physical servers: test one host per detector

BRIDGE-HOST PROPOSAL

2 active hosts total

One for VD, one for HD.

Dedicated failover partners would make four.

STILL TO VALIDATE

Full production load

PTCs, other interfaces, full namespaces, native latency, Ignition/historian load and failover.

[L1] A production minimum is unmeasured. The populations ran separately on one test host. Ignition, database and instrument PCs require separate sizing.

Run it: open a terminal in your sc_emu checkout

```
bash tools/setup.sh
.venv/bin/python tools/run.py smoke
```

REQUIREMENTS

Linux/WSL, Python 3.13, internet for setup

Use the updated checkout containing `tools/setup.sh`. The installation creates `.venv` in that checkout.

SMALL CHECK

Wait for PASS: 2 topologies completed

Uses 6 WIBs + 2 DAPHNEs; no detector hardware needed.

[R1] Run all following commands from this repository directory, wherever you keep it.

Run VD, then HD, then open the reports

```
.venv/bin/python tools/run.py vd
.venv/bin/python tools/run.py hd
less runs/latest-vd/report.md
less runs/latest-hd/report.md
```

EACH RUN

Wait for PASS: 6 topologies completed

Allow several minutes. In the report viewer, use arrow keys to scroll and q to quit.

[R1] Keep the timestamped run directory: report, inventory, provenance and package versions. Timings vary by computer.

Optional: rerun the configuration projection

```
.venv/bin/python tools/project_wib_queues.py
```

OUTPUT

Open report.md in the printed run directory

Run after VD and HD finish. Full instructions: `sc_emu/RUNBOOK.md`.

Report term	Meaning
Values/s	Completed OPC-UA Read values per second
RSS	Server-process memory
p95	95% of observed or modeled times are at or below this value

[Q1, R1] Raw reports and simulation code belong to sc_emu. Source links and assumptions: SOURCES.md.